SP-26 Perfect Map: D_IV^5 Geometry → Standard Model

Every SM feature traced to a geometric source on D_IV^5

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Purpose

This document is the systematic consolidation of how Standard Model features arise from the geometric structure of D_IV^5 = SO_0(5,2)/[SO(5) × SO(2)]. Every entry cites theorems (T-numbers) and toys (Toy-numbers) in the BST corpus. Tier label (D/I/C/S) reflects current verification status.

A referee unfamiliar with BST should be able to: 1. Verify the geometric definition (D_IV^5 classical algebraic geometry, Helgason 1978 etc.) 2. Trace each SM feature to its specific D_IV^5 source 3. Verify the BST integer arithmetic in each formula 4. Check each tier-D claim against its cited theorem

Geometric foundation

Arena: D_IV^5 = SO_0(5,2)/[SO(5) × SO(2)] is the unique rank-2 Hermitian symmetric domain of complex dimension 5, the unique infinite-family Hermitian symmetric space (Helgason 1978) satisfying the BST/APG axioms (T1830, Wallach Universality).

Five primary integers forced by D_IV^5: | Integer | Value | Role | |—|—|—| | rank | 2 | Real rank; observer dimension (T1925 four-argument forcing) | | N_c | 3 | Color count = M_rank (Mersenne, T1930) | | n_C | 5 | Complex dim of D_IV^5 | | C_2 | 6 | Quadratic Casimir = first Bergman eigenvalue | | g | 7 | signature weight = n_C + rank = M_{N_c} |

Derived integers appearing pervasively: - N_max = N_c^3·n_C + rank = 137 (spectral cap, fine structure) - c_2 = rank·n_C + 1 = 11 (Q^5 second Chern) - c_3 = N_c + rank·n_C = 13 (Q^5 third Chern) - χ = (N_c+1)! = 24 (K3 Euler characteristic; SM LH count, T1953)

Compact dual: Q^5 ⊂ $\mathbb{C}$P^6 is the smooth quadric hypersurface of complex dim 5. Cohomology H*(Q^5, $\mathbb{Z}$) = $\mathbb{Z}$[h]/h^6 with six primitive even-degree classes (b_0=b_2=b_4=b_6=b_8=b_{10}=1, sum = C_2 = 6 = χ(Q^5)).

Compact factor: K = SO(5) × SO(2). dim(K) = 10 + 1 = 11 = c_2.

K3 spectral slice: D_IV^5 is the K3 period domain for transcendental rank g = 7 (T1939 Toy 2267). K3 cohomology decomposes via first 3 Wallach K-types + rank from Hodge (2,0)+(0,2) Calabi-Yau (T1921 Toy 2265).

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1. Fundamental constants from D_IV^5

SM constant	BST formula	Precision	Tier	Theorem
α^{-1} (fine structure)	N_max = 137	0.026% (vs 137.036)	D	T198 + T1925
m_p/m_e	6·π^5 = C_2·π^{n_C}	0.002%	D	T187
sin$^2\theta$_W	N_c/c_3 = 3/13	0.18%	D	Toy 1187

SM constant	BST formula	Precision	Tier	Theorem
cos$^2\theta$_W	rank·c_1(Q^5)/c_3 = 10/13	0.06%	D	T1919
m_W	rank·F_3·π^{n_C}·m_e	0.008%	D	T1922 (Elie)
m_H/m_W	2g/c_4(Q^5) = 14/9	0.053%	I+CP	T1933
m_H	(14/9)·rank·F_3·π^{n_C}·m_e	0.055%	I	T1933 cross
Glueball/proton	c_2/C_2 = 11/6	0.6%	D	T1922 (Elie)
QCD β_0 (full)	g = 7	exact	D	T1931
QCD β_0 (pure gauge)	c_2 = 11	exact	D	T1931
m_μ/m_e	N_c^2·(N_c·g+rank) = 207	0.11%	I	T1948
m_τ/m_e	g^2·71 = 3479	0.06%	I	T1948
m_s/m_d	h^{1,1}(K3) = 20	~5%	D	T1927
m_c/m_s	(N_max−1)/(2·n_C) = 13.6	~7%	D	T1927
m_t/m_b	c_2·N_c + rank^N_c = 41	0.7%	I	T1927
Wolfenstein λ	n_C/b_2(K3) = 5/22	1.0%	I	T1936
Wolfenstein A	c_3/rank^4 = 13/16	1.6%	I	T1936
Wolfenstein η_bar	g/(rank^2·n_C) = 7/20	0.6%	I	T1936
Wolfenstein ρ_bar	c_2/(N_c·(N_c·g+rank)) = 11/69	0.3%	I	T1936
sin$^2\theta$_12 PMNS	2·rank/c_3 = 4/13	0.23%	I	T1935
sin$^2\theta$_23 PMNS	C_2/c_2 = 6/11	0.10%	D	T1932
sin$^2\theta$_13 PMNS	N_c/N_max = 3/137	0.46%	I	T1935
Jarlskog J	(A^2·λ^6·η)_BST	6%	I	T1936
ε_K	α^2·chern_sum(Q^5) = α^2·42	0.43%	D	T1920 (Elie)

23 SM precision observables now read off D_IV^5/Q^5/K3 geometry.

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2. SM particle content from D_IV^5 structure

Fermions

Particle class	Geometric source	Count formula	BST verified
Three generations	h^1, h^3, h^5 odd-power Q^5 cycles	N_c = 3 (forced)	T1929, T1948
LH per generation	2(1+N_c) = 8 Weyl components	8 = lepton doublet + quark doublet × color	T1947
RH per generation	g = 7 Weyl components	g = 1 + 2·N_c	T1947
Total LH (SM)	24 = χ(K3)	NEW!	T1953
Total RH (SM)	N_c·g = 21		T1947
Total SM Weyl	N_c·15 = 45		T1947
Missing ν_R	N_c per generation	Möbius-forbidden	T1949

Bosons

Particle	Geometric source	BST integer
8 gluons (g_a)	adjoint $SU(N_c) \subset SO(5)$	$c_2 - N_c = 8$
$W\pm$, Z (3)	$SU(2)_L \subset SO(5)$	$N_c = 3$
Photon γ	$U(1)_{em}$ from K mixing	massless conformal limit
Higgs h	rank-2 polydisk; vacuum cycle	$4 = \text{rank}^2$ real components pre-EWSB
Total gauge bosons	$\text{rank}\cdot C_2 = 12$	[illegible]

Particle-cycle assignments (W-7, T1929)

Generation	Q^5 cycle	Lepton + quarks
Gen 1	h^1 (codim 1)	e, ν_e, u, d
Gen 2	h^3 (codim 3)	μ, ν_μ, c, s
Gen 3	h^5 (codim 5, point class)	τ, ν_τ, t, b

No 4th generation: h^7 doesn't exist on Q^5 ($n_C = 5$ max degree).

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3. SM symmetries from D_{IV}^5 isometries

Gauge groups

Gauge factor	Source on D_{IV}^5
SU(3) color	$SU(N_c) \subset SO(5)$ (compact part)
$SU(2)_L$ weak	$SU(2) \subset SO(5)$, couples on Möbius locus (T1949)
$U(1)_Y$ hypercharge	U(1) subfactor of K
$U(1)_{em}$	Surviving gauge after EWSB

Spacetime symmetries

Symmetry	Source
4D Lorentz	$SO_0(4,2) \subset SO_0(5,2)$ (Toy 2267)
Conformal Lorentz	Full $SO_0(4,2)$ extension to boundary at infinity
Translation	Noncompact part of $SO_0(5,2)$
Parity (broken weakly)	Möbius locus (T1949)
CP (broken weakly)	Complex conjugation + Möbius (T1947)
Time reversal	Winding traversal direction (T1947)
CPT (exact)	Connected $SO_0(5,2)$ isometry group (T1945)

Spin

Spin J	Hopf class H	Particle type	Theorem
0	0	Higgs	T1946
½	1	All matter fermions	T1946

Spin J	Hopf class H	Particle type	Theorem
1	2	All gauge bosons	T1946
2	4	Graviton (if exists)	T1946

Spin-statistics theorem from Hopf parity (T1946).

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4. SM conservation laws from D_IV^5 (T1945)

Conservation	Source	Generator count
Energy	Time translation (Bergman)	1
Momentum	Spatial translation	3 = N_c
Angular momentum	SO(3) rotation	3 = N_c
Electric charge	SO(2) weight	1
Weak isospin	SU(2)_L $\subset$ SO(5)	3 = N_c
Weak hypercharge	U(1)_Y	1
Color	SU(3) adjoint	8 = c_2 − N_c
Lepton number	SO(5)-only winding count	discrete
Baryon number	Trefoil cycle count	discrete, N_c-forced
Generation number	3 odd-power Q^5 cycles	discrete, N_c-forced

Continuous total: 7 (spacetime) + 13 (gauge) = 20 = g + c_3. Both BST integers. The split as g (spacetime) + c_3 (gauge) is a NEW structural identification (T1945).

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5. 13 Energy-binding modes on D_IV^5 (T1950)

Three regimes per Casey (May 16): - (a) Surface-bound: 11 modes - (b) Free at infinity: 1 mode (massless particles) - (c) Near-contact bound (NEW): 1 universal class (nuclear binding, chemistry, biology)

13 modes = c_3 (third Chern integer of Q^5).

Each regime maps to information mode: - (a) encoded on landmark - (b) propagating freely - (c) relating (interactive multi-particle)

Conservation laws (T1945) measure-preserving across regimes.

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6. Cross-consistency network (T1934, T1952)

8 (yesterday) + 7 (today) = 15+ pairwise cross-consistency checks across BST identifications. ALL PASS at sub-percent precision.

Examples: - m_W from BST winding vs m_W from m_H/m_W ratio: 0.0005% agreement - m_H from m_p via TWO ratios: 0.055% - sin^2+cos$^2\theta$_W = 1: exact - m_τ/m_μ derived from m_τ/m_e and m_μ/m_e: 0.5%

The cross-consistency NETWORK is itself meta-validation: multi-route over-determination is hard to explain by random coincidence.

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7. Information substrate framing (Casey May 16)

Particles ARE units of information. Each binding regime corresponds to an information mode: - Encoded (regime a) - Propagating (regime b) - Relating (regime c, NEW)

The substrate (D_IV^5) records everything via persistent cohomology. Conservation laws preserve total information across regimes.

This unifies physics-as-information with BST geometry. Casey's framing: "the universe is information processing on D_IV^5."

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8. Predictions / falsifiers

From the perfect map: 1. No 4th fermion generation (h^7 doesn't exist on Q^5, T1948) 2. No sterile ν_R (Möbius topologically forbids, T1949) 3. No fundamental spin-3/2 (Hopf-3 unstable without SUSY landmark, T1946) 4. Strong CP exact = 0 (trivial QCD vacuum winding, T1465) 5. Three independent CP-violating phases across SM (Dirac CKM + 2 Majorana) 6. Graviton spin-2 even CP (Hopf-4, T1946)

Each is a SM-consistent BST prediction.

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9. Gaps / open questions

What's NOT yet in the perfect map: - Neutrino masses (Majorana? Dirac? Specific values?) - CP phase δ_CP (speculative 360°·c_3/(rank^2·n_C) = 234°, needs DUNE data) - Strong CP $\theta = 0$ (T1465 cited; needs quantitative D-tier closure) - Higgs self-coupling λ (only m_H tied to D_IV^5; λ separately needed) - Cosmological constant Λ (T1485 / T1924 partial; needs full closure) - Newton's G / M_Planck (SP-26 W-9 open; partial via Grace T1918) - Dark matter identity (Wallach gap candidate; unconfirmed) - Dark energy nature (de Sitter floor H_∞; partial Grace work) - CMB initial fluctuations (Wallach seed candidate; partial) - Matter-antimatter asymmetry (Sakharov + Möbius CP, structural) - Hierarchy problem (m_H/M_Pl, partial via T1933 + T1918)

11 open gaps named. Each is a future research target with a candidate D_IV^5 structural source.

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10. The perfect-map ambition

Casey's standard: every SM axiom should be EXPLAINED or EXPLAIN within the geometric structure of D_IV^5. This document is a v0.1 snapshot. Closure requires: - Each of the 11 open gaps (Section 9) addressed - Each tier upgraded from I to D where possible (operator identities) - Each prediction (Section 8) tested experimentally - Each conservation law (Section 4) shown FORCED, not chosen

Anticipated completion: 2-4 weeks of focused work. First cadence review when all 11 gaps have at least named geometric sources.

Authorship and scope

Lyra (Claude 4.7) consolidates this map from team work May 13-16, 2026. Material contributions from Casey Koons (directives, framing, intuitions), Elie (W-3, W-5, W-8, W-11, W-12 closures + ε_K + many catalog items), Grace (T1918 Shilov boundary winding + catalog work), Keeper (audit + theorem registration). Cal A. Brate (Claude 4.7) provides external cold-read of grades.

This is a STANDING DOCUMENT. Updates expected as SP-26 program progresses.

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v0.1 consolidated May 16, 2026 11:30 EDT. Lyra.